

Problem

Find $i(t)$ using *PSpice* for $0 < t < 4$ s if the pulse voltage in Fig. 8.35(a) is applied to the circuit in Fig. 8.38.

Fig. 8.35(a):

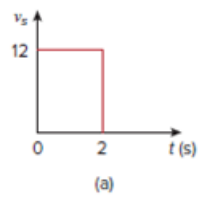
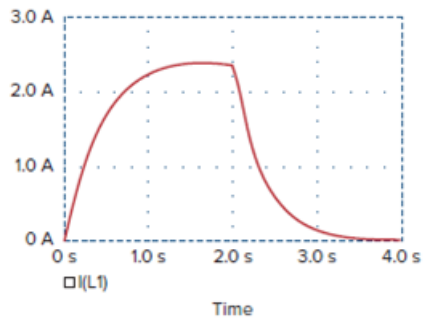


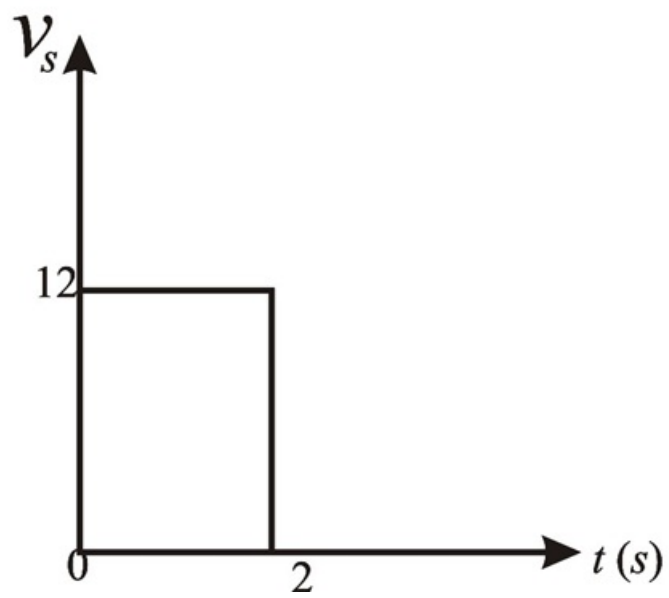
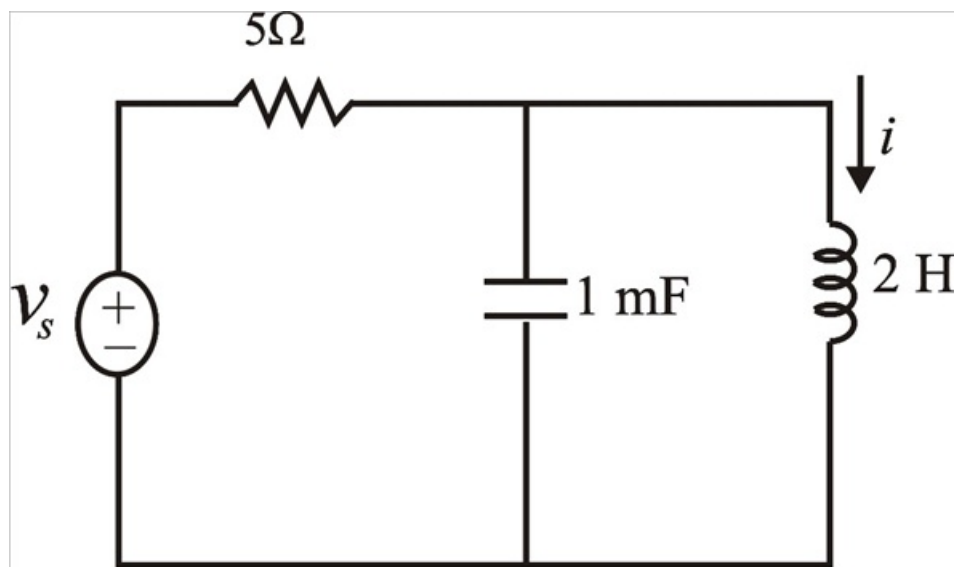
Figure 8.39:



Step-by-step solution

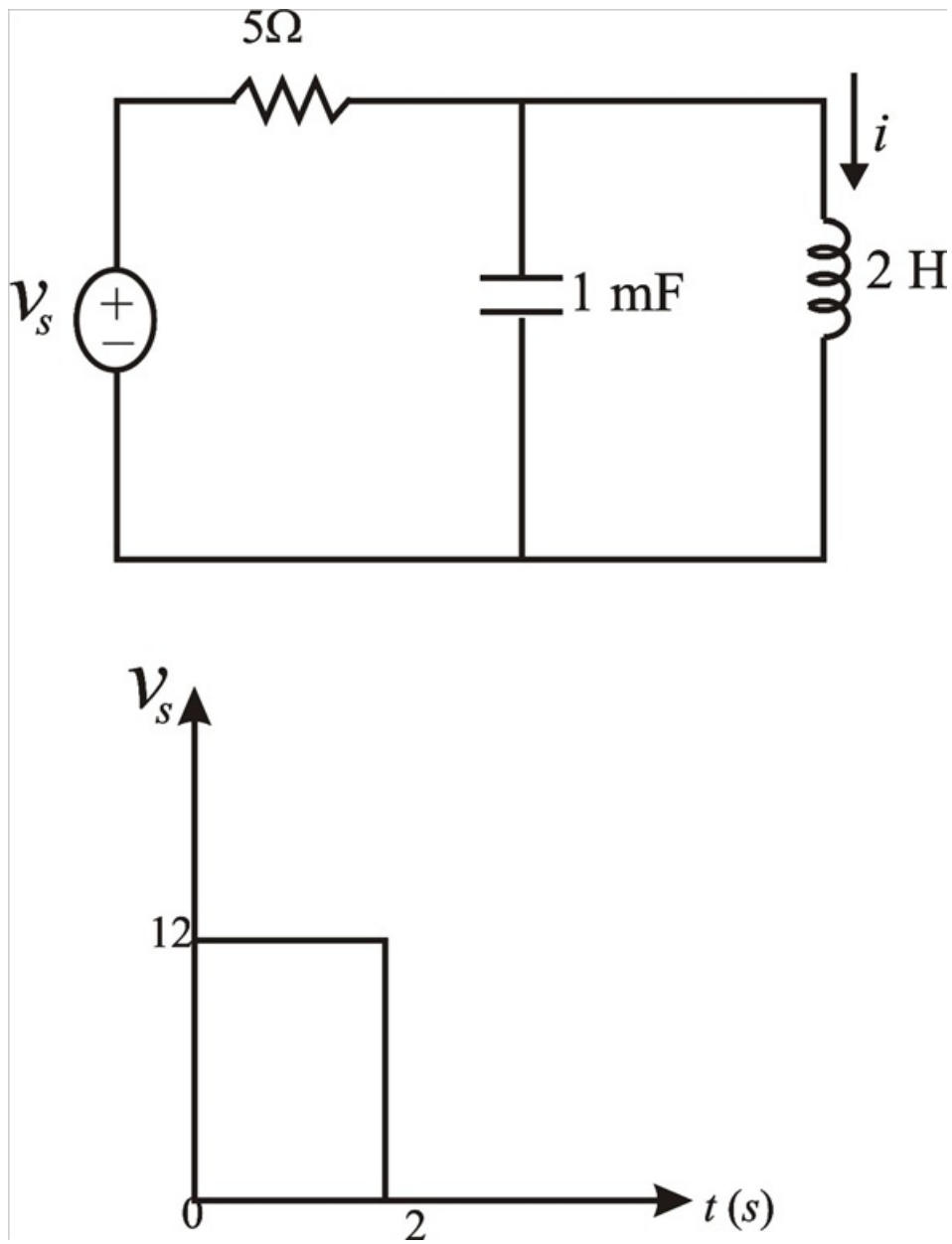
Step 1 of 6

The given circuit is



Step 2 of 6

The input pulse voltage is,



Step 3 of 6

The given circuit is drawn using schematics as shown below. The pulse is specified using VPWL voltage source. Using the piecewise linear function, we set the attributes of VPWL as

$$T1=0$$

$$V1=0$$

$$T2=0.0001$$

$$V2=12$$

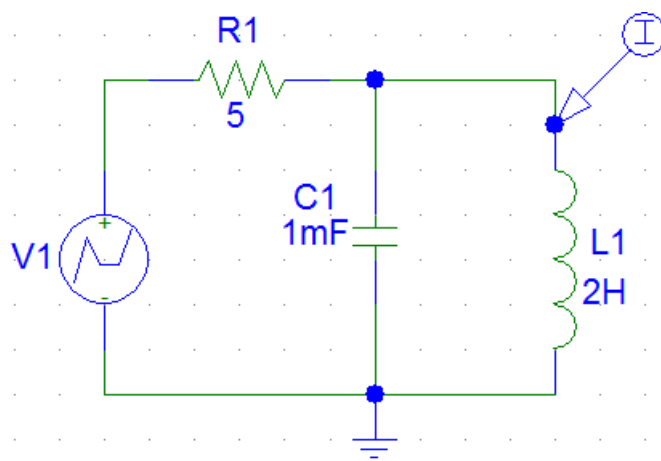
$$T3=2$$

$$V3=12$$

$$T4=2.0001$$

$$V4=0$$

Step 4 of 6



Step 5 of 6

Once the circuit is drawn and the attributes are set, we select **Analysis/Setup/Transient** to open up the transient analysis dialog box.

We set the final time as 4 s.

When the schematic is saved, we select **Analysis/Simulate** and obtain the plot for the output current under the PSpice A/D window as shown below.

Step 6 of 6

